LESSON PLAN

AIRCRAFT ARMAMENT (18)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Brief description on application of Radar in surveillance operation.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Brief description on application of Radar in surveillance operation. Features generally available on surveillance radar. Information obtained by surveillance radar. Concept of controlling and supporting aircraft taking part in surveillance operation and methods of passing information.
5. Specific Objectives	18.1.4.1 Trainee will be able to appraise the function of secondary radars in ASW operation.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of surveillance radar features and information flow diagrams using PPT slides.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a surveillance radar?
(b) Name two types of radar used in ASW operations.
(c) How does a secondary radar differ from a primary radar?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Application of Radar in Surveillance • Features of Surveillance Radar • Information Obtained by Surveillance Radar	L & Ds	PPT, IB	15 min
(b)	• Concept of Controlling Aircraft • Supporting Aircraft in Surveillance • Methods of Passing Information	L & Ds	PPT, PDF	20 min
(c)	• Function of Secondary Radars in ASW • Data Link Protocols • Surveillance Data Integration	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of secondary radar in Anti-Submarine Warfare operations.
(b) Describe the standard methods used for passing surveillance information between aircraft.

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts learned in this lesson apply to the basic concept of ESM covered in LP 10.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Basic concept of ESM. Types of Information extracted like PRF, PRT, and PW etc. and their utilization for verifying types of Radar.	Refer to the master syllabus reference list	LP 10

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min